Worldview OS

Technical Build Ticket Breakdown

Execution-grade engineering backlog for MVP delivery. Organized by epics, tickets, dependencies, acceptance criteria, and recommended sequencing.

Build goal: ship a usable MVP that takes a user question, normalizes it, routes the right lenses, generates scenarios, scores probabilities and impact, and returns a structured synthesis with memory and source-backed evidence.

Recommended MVP stack: Next.js frontend, Python/FastAPI orchestration backend, Postgres, pgvector, Redis queue, object storage, LLM provider abstraction, news/data adapters, and audit logging.

Planning assumption: one product-minded engineer, one full-stack/backend engineer, one frontend engineer, and part-time PM/design support over 8 weeks.

How to use this document

Each epic is a delivery track. Each ticket is small enough to estimate, assign, and verify.

Priority uses P0, P1, and P2. P0 is required for MVP launch.

Acceptance criteria are written so engineering and product can both sign off without philosophical debates.

Release scope at a glance

Phase	Scope	Priority	Exit condition
Foundation	Auth, app shell, orchestration service, data model, logging	P0	Core stack deployed and environments stable
Analysis MVP	Question normalize, lens routing, evidence retrieval, scenario engine, synthesis	P0	User can ask a question and receive structured output
Memory + Watch	Saved analyses,	P1	Analyses persist and

	forecast history, watchlist and indicator tracking		can be revisited
Admin + Hardening	Prompt config, evaluation, analytics, throttling, policy guardrails	P1	System observable, governable, and safer to operate

Epic 1. Platform Foundation

Create the application shell, environments, auth, persistent storage, and baseline developer workflows.

MVP priority: P0

ID	Ticket	Description	Owner	Depends on	Acceptance criteria
WF-001	Set up monorepo and CI/CD	Create frontend, backend, shared types, linting, tests, staging, and production deploy pipelines.	Platform Eng	None	Repo boots locally; CI runs lint and test; staging deploys automatically on main branch.
WF-002	Implement auth and user tenancy	Add email/social login, session handling, and user isolation for saved analyses and settings.	Backend	WF-001	User can sign in, sign out, and access only their own records.
WF-003	Create core database	Provision Postgres with	Backend	WF-001	Migration runs cleanly;

	schema	entities for users, questions, analyses, scenarios, evidence items, saved views, and watchlists.			entities query correctly; rollback tested.
WF-004	Add observability and audit logging	Capture request IDs, prompt versions, model calls, retrieval calls, token usage, and error traces.	Platform Eng	WF-001	Every analysis request produces a correlated audit trail visible in logs dashboard.
WF-005	Environment and secrets management	Set up environment variables, secret rotation, provider keys, and config profiles by environment.	Platform Eng	WF-001	No hardcoded secrets; staging and prod configs separated; secret retrieval documented.

Epic 2. Question Intake and Normalization

Turn loose user prompts into precise analysis jobs with defined targets, horizons, and domains.

MVP priority: P0

ID	Ticket	Description	Owner	Depends on	Acceptance criteria
WF-101	Build question intake API	Accept raw user question,	Backend	WF-003	API validates inputs and

		optional domain, time horizon, and objective.			stores incoming analysis request.
WF-102	Implement question normalization service	Rewrite vague questions into analyzable forms; extract success/failure criteria and ambiguous terms.	Backend/AI	WF-101	Service returns normalized question, definitions, clarifying assumptions, and unresolved ambiguities.
WF-103	Create domain classifier	Identify relevant domains such as politics, macro, markets, crypto, sports, gaming, or mixed-domain.	Backend/AI	WF-101	Classifier returns one or more domains with confidence score.
WF-104	Add user-editable normalization step in UI	Let user accept, tweak, or override normalized question and assumptions before full analysis.	Frontend	WF-101, WF-102	User can edit normalized question and rerun without losing original prompt.
WF-105	Persist prompt lineage	Store original question, normalized question, selected options, and	Backend	WF-102	History shows full lineage and rerun reproducibility.

		prompt template version.			

Epic 3. Lens Routing and Angle Library

Map each question to the correct analysis lenses and domain-specific angle packs.

MVP priority: P0

ID	Ticket	Description	Owner	Depends on	Acceptance criteria
WF-201	Create universal lens registry	Define universal lenses: structural, incentive, power, timing, historical, market, reflexivity, second-order, fragility, narrative, legal, behavioral, base-rate, tail-risk, and data.	Backend/AI	WF-003	Registry is queryable by service and editable by admin config.
WF-202	Create domain plug-in packs	Add domain-specific lenses for politics, markets, crypto, NBA, and racing games.	Backend/AI	WF-201	At least 5 domain packs exist and can merge with universal registry.
WF-203	Implement lens routing	Select and rank relevant	Backend/AI	WF-102, WF-103,	Engine returns ranked lenses

	engine	lenses for a given normalized question.		WF-201, WF-202	with explanation of why each lens matters.
WF-204	Add lens coverage guardrail	Detect missing critical lenses and force fallback expansion when coverage is weak.	Backend/AI	WF-203	Mixed-domain prompts do not return fewer than minimum required lenses.
WF-205	Build lens display card UI	Render selected lenses with reason, drivers, and confidence.	Frontend	WF-203	User sees clean lens list with expandable details and no duplicate lenses.

Epic 4. Evidence Retrieval and Source Handling

Collect source-backed evidence and opposing evidence before scenario generation.

MVP priority: P0

ID	Ticket	Description	Owner	Depends on	Acceptance criteria
WF-301	Create retrieval adapter interface	Define adapters for web/news, market data, crypto data, sports data, and internal memory retrieval.	Backend	WF-003	New adapters can be registered without changing orchestration core.
WF-302	Implement	Fetch recent	Backend	WF-301	System returns

	web/news retrieval adapter	and relevant sources with metadata, snippets, and links.			citations with recency and source attribution.
WF-303	Implement market and crypto data adapters	Connect price, macro, and optionally on-chain data providers.	Backend	WF-301	Time-series and point-in-time data can be attached to analysis jobs.
WF-304	Implement evidence ranking and contradiction pass	Rank supporting and counter-evidence; identify duplicated or low-quality evidence.	Backend/AI	WF-302, WF-303	At least one opposing-evidence bucket is created when applicable.
WF-305	Build evidence panel UI	Show evidence items, source links, and tags such as supporting, opposing, leading indicator, or disconfirming signal.	Frontend	WF-304	User can inspect evidence without losing current analysis context.

Epic 5. Scenario Engine and Scoring

Generate scenario trees, quantify outlooks, and expose what changes the score.

MVP priority: P0

ID	Ticket	Description	Owner	Depends on	Acceptance

					criteria
WF-401	Implement hypothesis generator	Create candidate hypotheses from lenses and evidence.	Backend/AI	WF-203, WF-304	System outputs coherent hypotheses tied to explicit lenses.
WF-402	Implement scenario generator	Produce base, bull, bear, and wildcard scenarios with descriptions.	Backend/AI	WF-401	Every analysis returns at least 3 scenarios; wildcard optional only when unsupported.
WF-403	Implement scoring model	Score probability, impact, time horizon, confidence, and score-change triggers.	Backend/AI	WF-402	Scores are present for each scenario and sum/format rules are enforced.
WF-404	Implement hidden variables and change-my-mind pass	Generate hidden variables, disconfirming signals, and what would change the model’s view.	Backend/AI	WF-402	Output includes underappreciated variables and explicit revision triggers.
WF-405	Build scenario matrix UI	Render sortable scenario matrix with scenario	Frontend	WF-403, WF-404	Matrix readable on desktop; rows expandable; saved state

		description, score fields, and indicators.			supported.

Epic 6. Synthesis and Final Output

Merge analysis into a clean, decision-grade answer rather than a wall of generated soup.

MVP priority: P0

ID	Ticket	Description	Owner	Depends on	Acceptance criteria
WF-501	Implement committee orchestration	Run specialist roles such as political analyst, macro strategist, market analyst, crypto analyst, historian, red team, forecaster, and synthesizer.	Backend/AI	WF-203, WF-304, WF-402	Workflow can run specialist prompts in sequence or parallel and preserve outputs.
WF-502	Implement synthesis composer	Generate final output sections: reframed question, definitions, lenses, drivers, scenario matrix, arguments for, arguments against, hidden variables,	Backend/AI	WF-501	Final answer contains required sections in stable order.

		change-my-mind, bottom line, and watchlist.			
WF-503	Add user objective action layer	Tailor closing section for investor, operator, researcher, content creator, or bettor use case.	Backend/AI	WF-502	Final answer differs meaningfully by selected objective.
WF-504	Build final analysis screen	Show synthesis, scenario matrix, evidence, and watchlist in one navigable experience.	Frontend	WF-405, WF-502	User can jump between sections and export analysis.
WF-505	Export analysis to PDF/Doc	Create shareable export with question, scores, evidence summary, and date stamp.	Backend/ Frontend	WF-504	User can download readable PDF or doc export.

Epic 7. Memory, Forecast History, and Watchlists

Make the system learn over time instead of resetting to zero context every session.

MVP priority: P1

ID	Ticket	Description	Owner	Depends on	Acceptance

					criteria
WF-601	Implement saved analyses and folders	Allow users to save, label, and organize outputs.	Backend/ Frontend	WF-003, WF-504	Saved analyses persist and open quickly.
WF-602	Implement forecast history schema	Store previous scenario scores and revisions over time.	Backend	WF-403	History records can be queried by topic and date.
WF-603	Build watchlist and indicator tracker	Let users watch a topic and track selected leading indicators.	Backend/ Frontend	WF-304, WF-403	User can follow a topic and see latest indicator changes.
WF-604	Add memory retrieval into new analyses	Pull related prior analyses and priors into new jobs when relevant.	Backend/AI	WF-602	Related history appears and influences synthesis with traceability.
WF-605	Build history and watchlist screens	Create screens for saved analyses, score change timeline, and tracked topics.	Frontend	WF-601, WF-603	User can scan history and reopen analysis snapshots.

Epic 8. Admin, Evaluation, and Safety

Give operators tools to manage prompts, measure quality, and avoid silent system decay.

MVP priority: P1

ID	Ticket	Description	Owner	Depends on	Acceptance criteria
WF-701	Create prompt configuration console	Version prompts, lens packs, model settings, and objective templates.	Backend/ Frontend	WF-001	Admin can change config without code deploy.
WF-702	Build evaluation harness	Run benchmark questions and compare output quality, coverage, and stability across prompt versions.	Backend/AI	WF-701	Benchmark report generated with pass/fail thresholds.
WF-703	Add hallucination and citation guardrails	Detect unsupported claims, missing sources, and stale data use on current-event prompts.	Backend/AI	WF-302, WF-502	System flags unsupported claims before response finalization.
WF-704	Usage analytics and cost controls	Track model costs, latency, abandonment, save rate, and feature usage.	Backend/ Frontend	WF-004	Dashboard shows per-feature usage and cost trends.
WF-705	Rate limiting and abuse controls	Throttle heavy users, protect APIs, and prevent runaway loops.	Platform Eng	WF-001	Load tests pass; requests are constrained gracefully under abuse

conditions.

Recommended implementation sequence

Sprint	Goal	Primary tickets	Definition of done
1	Foundation live	WF-001 to WF-005	App runs in staging with auth, DB, logging, and secure config
2	Question intake	WF-101 to WF-105	User submits question; system normalizes and stores lineage
3	Lens routing	WF-201 to WF-205	System selects lenses and shows them in UI
4	Evidence retrieval	WF-301 to WF-305	Evidence and opposing evidence appear with citations
5	Scenario engine	WF-401 to WF-405	Scenario matrix generated with score fields and indicators
6	Synthesis	WF-501 to WF-505	Committee flow and final analysis screen usable end to end
7	Memory	WF-601 to WF-605	Saved analyses, history, and watchlists work
8	Admin + hardening	WF-701 to WF-705	Prompt config, evaluation, analytics, and safety controls active

Definition of MVP completion

- A signed-in user can submit a question, review the normalized version, run analysis, inspect lenses, review evidence, view a scored scenario matrix, and read a structured synthesis.
- Every output has traceable prompt lineage, source attribution where applicable, and audit logs for debugging.
- At least one saved analysis flow and one watchlist flow work end to end.
- Latency, cost, and error tracking are visible to operators.

Notes for engineering estimation

Treat AI-heavy tickets as thin vertical slices first. A mediocre but testable end-to-end path beats a clever architecture graveyard.

Do not let the retrieval layer sprawl. MVP only needs adapters that materially improve answer quality for your target domains.

Prompt versioning and audit logging are not optional. Without them, debugging this system will be a clown show.